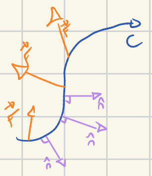
Lecture 23: Flux

Flux: Another kind of line integral

C: plane curve
$\vec{F}$: vector field

Flux of $\vec{F}$ across C is: $\boxed{\int_C \vec{F}\cdot\hat{n}\,ds}$



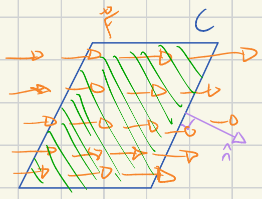
$\hat{n}$: unit normal vector to C, $90°$ clockwise from $\hat{T}$ (tangent vector)

If we break C into small pieces Δs: Flux $= \lim_{\Delta s\to 0}\left(\sum \vec{F}\cdot\hat{n}\,\Delta s\right)$

Work: $\int_C \vec{F}\cdot d\vec{r} = \int_C \vec{F}\cdot\hat{T}\,ds$ Summing tangential component of $\vec{F}$

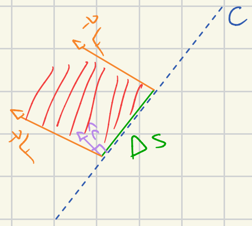
Flux: $\int_C \vec{F}\cdot\hat{n}\,ds$ Summing normal component of $\vec{F}$

Interpretation: for $\vec{F}$ a velocity field, Flux measures how much fluid passes through C per unit time.



▨ : amount of fluid passing through C per unit time

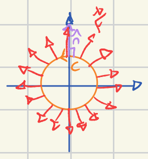
What passes through a portion of C in unit time?
Area = base · height = $\Delta s\cdot(\vec{F}\cdot\hat{n})$



What flows across C (left-to-right) is counted positively
(right-to-left) is counted negatively

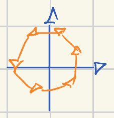
Example: C = circle of radius a, centered at the origin, counterclockwise

$\vec{F} = x\hat{\imath} + y\hat{\jmath}$



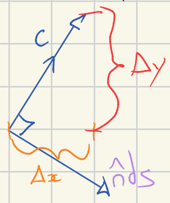
Along C, $\vec{F} \parallel \hat{n} \Leftrightarrow \vec{F}\cdot\hat{n} = |\vec{F}|\,\cancel{|\hat{n}|}^{=1} = a$ $\quad \int_C \vec{F}\cdot\hat{n}\,ds = \int_C a\,ds = a\cdot\underbrace{\text{length}(C)}_{2\pi a} = 2\pi a^2$

Same C, $\langle -y, x\rangle$



$\vec{F}$ tangent to C so $\vec{F}\cdot\hat{n} = 0$, therefore Flux $= 0$

* Calculation using components: Remember: $d\vec{r} = \hat{T}ds = \langle dx, dy\rangle$ $\quad \hat{n}$ is $\hat{T}$ rotated $90°$ clockwise, $\boxed{\hat{n}\,ds = \langle dy, -dx\rangle}$



$\hat{T}\Delta s = \Delta\vec{r} = \langle \Delta x, \Delta y\rangle$
$\hat{n}\,ds = \langle \Delta y, -\Delta x\rangle$

So, if $\vec{F} = \langle P, Q\rangle$ then $\int_C \vec{F}\cdot\hat{n}\,ds = \int_C \langle P, Q\rangle\cdot\langle dy, -dx\rangle = \int_C -Q\,dx + P\,dy$

If $\vec{F} = \langle M, N\rangle$: $\int_C \vec{F}\cdot\hat{n}\,ds = \int_C -N\,dx + M\,dy$

Green's theorem for flux: If C encloses a region R counterclockwise and $\vec{F} = \langle P, Q\rangle$ defined in R, then $\oint_C \vec{F}\cdot\hat{n}\,ds = \iint_R \underbrace{\text{div}(\vec{F})}_{\text{divergence of }\vec{F}}\,dA$

$\text{div}\langle P, Q\rangle = P_x + Q_y$ $\quad$ This is Green's theorem in normal form (vs Green's theorem in tangential form)

$$\oint_C \underbrace{-Q}_{M}\,dx + \underbrace{P}_{N}\,dy = \iint_R \underbrace{(P_x + Q_y)}_{N_x - M_y}\,dA \qquad \text{Let } M = -Q,\ N = P \qquad \oint M\,dx + N\,dy = \iint_R (N_x - M_y)\,dA$$

Example: $\vec{F} = x\hat{\imath} + y\hat{\jmath}$, C = circle of radius a $\quad \text{div}\,\vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) = 1 + 1 = 2$ $\quad \oint_C \vec{F}\cdot\hat{n}\,ds = \iint_R 2\,dA = 2\,\text{area}(R) = 2\pi a^2$

Physical Interpretation: in an incompressible fluid flow, divergence measures source/sink density/rate, i.e., how much fluid is being added to the system per unit area and unit time